



Final MISO DPP 2018 April South Area Study Phase I Report

August 15th, 2019

**MISO
720 City Center
Drive Carmel
Indiana 46032**
<http://www.misoenergy.org>



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1. Executive Summary

This report presents the results of a System Impact Study performed to evaluate the interconnection of the generators in the DPP 2018 April South Area Phase I (South Area DPP 1). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Entergy, Cooperative Energy, CLECO, AECL, Southern Company, PowerSouth, SPP, and TVA.

1.1. Project List

The original interconnection requests for DPP 2018 April South Area had a total of 49 generation projects. Projects J1029, J1133, J1134, J1136, and J1167 withdrew prior to the start of DPP Phase I. Therefore, there are 44 generation projects with a combined nameplate rating of 7,808 MW of Energy Resource Interconnection Service (ERIS) & 7,668 MW of Network Resource Interconnection Service (NRIS). The detailed list of South Area DPP 1 is shown below in Table 1, and the Phase I study was kicked off on March 27th, 2019.

Table 1: List of DPP 2018 April South Area Phase I Projects

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	Point of Interconnection (POI)
J1007	Solar	EES	Crittenden	AR	NRIS	200	Marion - Marked Tree 161 kV Line
J1020	Solar	CE	Jasper	MS	NRIS	250	Missionary 230 kV Substation
J1023	Solar	EES	Mississippi	AR	NRIS	82	Osceola - Wilson 161 kV Line
J1030	Solar	EES	Liberty	TX	ERIS	150	Gulf Livingston - Menard 138 kV Line
J1049	Solar	CE	Forrest	MS	NRIS	250	Purvis 161 kV Substation
J1060	Solar	EES	St. Francis	AR	NRIS	500	West Memphis - Keo 500 kV Line
J1070	Solar	EES	Washington	MS	NRIS	100	SE Greenville 115 kV Substation
J1073	Solar	EES	Iberville Parish	LA	NRIS	300	Webre 500 kV Substation
J1076	Solar	EES	St. James Parish	LA	NRIS	200	Vacherie 230 kV Substation
J1078	Solar	EES	Mississippi	AR	NRIS	300	San Souci 500 kV Substation
J1079	Solar	EES	St. Charles Parish	LA	NRIS	200	Waterford 230 kV Substation
J1091	Solar	EES	Craighead	AR	NRIS	150	Power Line Road 161 kV Substation
J1093	Solar	EES	Desha/Drew	AR	NRIS	200	Reed 115 kV Substation
J1095	Solar	EES	DeSoto	MS	NRIS	300	Twinkletown 230 kV Substation
J1097	Solar	EES	Lonoke/Pulaski	AR	NRIS	500	KEO 500 kV Substation
J1099	Solar	EES	Iberville Parish	LA	NRIS	200	Derrick 230 kV Substation
J1100	Solar	EES	Grimes	TX	NRIS	200	Grimes 345 kV Substation
J1113	Solar	EES	Sunflower	MS	NRIS	85	Delta - Ruleville 115 kV Line
J1120	Wind	EES	Jefferson	TX	NRIS	145	Big Hill - Bayou Farms 138 kV Line
J1125	Solar	EES	Cross	AR	NRIS	135	Wynne South - AECC Wynne E 161 kV Line
J1130	Gas	EES	Warren	MS	NRIS	1110	Baxter Wilson 500/115 kV Substation
J1137	Solar	EES	Jackson	AR	NRIS	100	Swifton - Hoxie South 161 kV Line
J1142	Solar	EES	St. James Parish	LA	NRIS	20	Vacherie 230 kV Substation
J1144	Solar	CE	Covington/Lamar	MS	NRIS	96	Moselle - Columbia 161 kV Line
J1146	Coal	EES	Mississippi	AR	NRIS	10	San Souci 500 kV Substation
J1147	Solar	EES	St. Francis	AR	NRIS	100	Moses - McCrory 161 kV Line
J1151	Gas	EES	Ascension Parish	LA	NRIS	219	Brittany 230 kV Substation
J1155	Solar	EES	Mississippi	AR	NRIS	150	Driver 230 kV Substation
J1156	Solar	EES	Crittenden	AR	NRIS	150	Kuhn Rd. - Parkin 161 kV Line
J1157	Solar	EES	Hinds	MS	NRIS	100	Tinnin Road 230 kV Substation
J1158	Solar	EES	St. James Parish	LA	NRIS	70	Vacherie 230 kV Substation
J1159	Solar	EES	Walker	TX	NRIS	100	Mount Zion 138 kV Substation
J1162	Solar	EES	Jefferson	TX	NRIS	80	China 138 kV Substation
J1163	Solar	EES	Quitman/Tunica	MS	NRIS	120	Crenshaw - Tunica 115 kV Line

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	Point of Interconnection (POI)
J1165	Solar	EES	Jefferson Davis Parish	LA	NRIS	150	LC Bulk - Henning 138 kV Line
J1166	Solar	EES	Acadia Parish	LA	NRIS	50	Bayou Cove - Richard 138 kV Line
J1168	Solar	EES	Liberty	TX	NRIS	80	Stilson 138 kV Substation
J1184	Solar	EES	Washington Parish	LA	NRIS	50	Lagen Holton - Franklinton 115 kV Line
J1186	Solar	EES	San Jacinto	TX	NRIS	100	Pelican Road - Shepard 138 kV Line
J961	Solar	EES	West Baton Rouge Parish	LA	NRIS	150	Livonia - Wilbert 138 kV Line
J962	Solar	EES	DeSoto	MS	NRIS	100	Twinkletown 230 kV Substation
J964	Solar	EES	Hinds	MS	NRIS	70	Edwards - Bolton 115 kV Line
J972	Solar	EES	St. Francis	AR	NRIS	140	Hughes 161 kV Substation
J988	Solar	CE	Wayne	MS	NRIS	200	W Waynesboro 230 kV Substation

1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2018 April South Phase I reflects responsibilities for mitigating system impacts. The total cost of Network Upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies. Details pertaining to the cost allocation has been detailed in Appendix A – Cost Allocation Summary (CEII).

Table 2: Total Cost of Network Upgrades for DPP 2018 April South Phase I Projects

Project	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)
	Thermal	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned				
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i	\$4,000/MW	10% of (j) from Phase I-M2
J1007	7,187,818	TBD	TBD	TBD	5,016,050	14,000,000	0	291,841	26,495,710	800,000	1,849,571
J1020	6,398,939	TBD	TBD	TBD	8,318,392	2,738,750	0	0	17,456,081	1,000,000	745,608
J1023	28,419	TBD	TBD	TBD	1,821,436	14,000,000	0	0	15,849,855	328,000	1,256,986
J1030	2,883,063	TBD	TBD	TBD	0	14,000,000	0	0	16,883,063	600,000	1,088,306
J1049	118,403,237	TBD	TBD	TBD	7,512,670	2,458,750	0	0	128,374,657	1,000,000	11,837,466
J1060	7,612,132	TBD	TBD	TBD	1,108,685	25,000,000	0	238,219	33,959,036	2,000,000	1,395,904
J1070	0	TBD	TBD	TBD	35,094,843	10,000,000	0	0	45,094,843	400,000	4,109,484
J1073	113,134,011	TBD	TBD	TBD	63,133,434	25,000,000	0	0	201,267,446	1,200,000	18,926,745
J1076	506,919	TBD	TBD	TBD	846,878	6,500,000	0	0	7,853,797	800,000	0
J1078	8,122,684	TBD	TBD	TBD	3,828,414	10,000,000	0	0	21,951,098	1,200,000	995,110
J1079	1,118,358	TBD	TBD	TBD	1,272,026	7,500,000	0	0	9,890,384	800,000	189,038
J1091	2,455,943	TBD	TBD	TBD	1,522,897	10,000,000	0	0	13,978,840	600,000	797,884
J1093	49,549,096	TBD	TBD	TBD	10,403,905	10,000,000	0	0	69,953,001	800,000	6,195,300
J1095	18,549,280	TBD	TBD	TBD	14,623,505	5,000,000	0	0	38,172,785	1,200,000	2,617,279
J1097	15,656,829	TBD	TBD	TBD	1,081,084	10,000,000	0	0	26,737,913	2,000,000	673,791
J1099	1,056,374	TBD	TBD	TBD	24,560,000	7,000,000	0	0	32,616,374	800,000	2,461,637
J1100	27,753,605	TBD	TBD	TBD	144,731,218	7,000,000	0	0	179,484,823	800,000	17,148,482
J1113	32,806,942	TBD	TBD	TBD	71,413,311	13,000,000	0	0	117,220,254	400,000	11,322,025
J1120	0	TBD	TBD	TBD	117,531,678	14,000,000	0	0	131,531,678	600,000	12,553,168
J1125	2,123,142	TBD	TBD	TBD	19,186,908	14,000,000	0	414,813	35,724,863	540,000	3,032,486
J1130	299,329,694	TBD	TBD	TBD	28,816,951	43,000,000	0	0	371,146,645	4,400,000	32,714,664
J1137	5,135,991	TBD	TBD	TBD	1,275,892	14,000,000	0	0	20,411,883	400,000	1,641,188
J1142	35,931	TBD	TBD	TBD	84,688	416,982	3,886,018	0	537,601	80,000	0
J1144	600,000	TBD	TBD	TBD	3,121,563	3,600,000	0	0	7,321,563	384,000	348,156
J1146	298,245	TBD	TBD	TBD	127,614	0	0	0	425,859	40,000	2,586
J1147	239,273	TBD	TBD	TBD	26,627,422	14,000,000	0	0	40,866,694	400,000	3,686,669
J1151	1,758,813	TBD	TBD	TBD	0	6,500,000	0	0	8,258,813	1,020,000	0
J1155	4,207,470	TBD	TBD	TBD	1,917,025	6,500,000	0	0	12,624,495	600,000	662,449
J1156	20,093,275	TBD	TBD	TBD	15,426,234	14,000,000	0	1,574,367	51,093,876	600,000	4,509,388
J1157	50,449,630	TBD	TBD	TBD	5,066,396	6,500,000	0	0	62,016,026	400,000	5,801,603

Project	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)
	Thermal	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned				
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i	\$4,000/MW	10% of (j) from Phase I-M2
J1158	147,612	TBD	TBD	TBD	296,407	6,500,000	0	0	6,944,019	280,000	414,402
J1159	7,620,000	TBD	TBD	TBD	73,506,095	13,000,000	0	0	94,126,095	400,000	9,012,610
J1162	0	TBD	TBD	TBD	63,979,750	17,500,000	0	0	81,479,750	320,000	7,827,975
J1163	18,213,005	TBD	TBD	TBD	4,145,983	13,000,000	0	0	35,358,988	480,000	3,055,899
J1165	0	TBD	TBD	TBD	160,641,799	14,000,000	0	5,053,846	179,695,645	600,000	17,369,565
J1166	0	TBD	TBD	TBD	35,236,454	14,000,000	0	0	49,236,454	200,000	4,723,645
J1168	22,008,736	TBD	TBD	TBD	61,340,177	4,000,000	0	0	87,348,914	320,000	8,414,891
J1184	175,455	TBD	TBD	TBD	41,807	2,377,000	91,000	0	2,594,262	200,000	0
J1186	16,486,937	TBD	TBD	TBD	74,620,701	14,000,000	0	0	105,107,638	400,000	10,110,764
J961	9,023,363	TBD	TBD	TBD	8,451,318	4,000,000	0	0	21,474,681	600,000	1,547,468
J962	7,774,433	TBD	TBD	TBD	4,874,502	5,000,000	0	0	17,648,934	400,000	1,364,893
J964	39,962,791	TBD	TBD	TBD	3,755,940	13,000,000	0	0	56,718,731	300,000	5,371,873
J972	5,302,555	TBD	TBD	TBD	30,192,833	14,000,000	0	0	49,495,388	600,000	4,349,539
J988	0	TBD	TBD	TBD	6,936,114	1,776,250	0	0	8,712,364	800,000	71,236

Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. ERIS Network Upgrades and NRIS Network Upgrades are shown in Table 3. Shared Network Upgrades are shown in Table 4.

Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Acadia - Scanlan 138 kV Rebuild	EES		J1165, J1166	4,152,000
Addis - Tiger 230 kV Rebuild	EES		J1099	24,560,000
Arcola - Hollandale 115 kV Rebuild	EES		J1070	12,886,000
Bayou Cove - Richard 138 kV New Third Line	EES		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	57,200,000
Bayou Labutte - Pt. Pleasant 230 kV New Line	EES		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	15,700,000
Bayou Labutte 500/230 kV Second Autotransformer	EES		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	18,000,000
Big Cajun - Coly 500 kV New Line	EES	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1100, J1113, J1120, J1130, J1144, J1146, J1155, J1157, J1159, J1162, J1163, J1165, J1166, J1168, J1186, J962, J964, J988	275,000,000
Canton - Pickens 230 kV Rebuild	EES		J1113	37,160,000
Champagne - Bobcat 138 kV Rebuild	EES		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	6,728,000
Coley Creek - Shepherd 138 kV Rebuild	EES	J1186	J1186	4,520,000
Colonial Academy - Acadia 138 kV Rebuild	EES		J1165, J1166	14,408,000
Coly - Willow Glen 500 kV Terminal Equipment Upgrade	EES		J1020, J1049, J1070, J1113, J1130, J1144, J1157, J1184, J964, J988	3,000,000
Danville 115 kV two 20 MVar Capacitor Banks	EES	J1007, J1023, J1060, J1073, J1076, J1078, J1079, J1091, J1093, J1095, J1097, J1099, J1125, J1137, J1142, J1147, J1151, J1155, J1156, J1158, J1163, J1184, J961, J962, J972		4,500,000
Franklin - Baxter Wilson 500 kV New Line	EES	J1007, J1020, J1049, J1060, J1073, J1078, J1091, J1093, J1095, J1097, J1100, J1113, J1125, J1130, J1137, J1146, J1155, J1156, J1157, J1163, J1168, J962, J964, J972		312,500,000
Franklin - Big Cajun 500 kV New Line	EES	J1007, J1020, J1049, J1060, J1073, J1078, J1091, J1093, J1095, J1097, J1100, J1113, J1125, J1130, J1137, J1146, J1155, J1156, J1157, J1163, J1168, J962, J964, J972		440,000,000
Greenville - Arcola 115 kV Rebuild	EES		J1070	13,096,000
Grimes - Mt. Zion 138 kV Rebuild	EES		J1100	27,125,000
Grimes 345/138 kV Third Autotransformer	EES		J1100	15,000,000
Hughes - West Memphis 161 kV Rebuild	EES		J1125, J1147, J1156, J972	28,000,000
Huntsville - Wyntx Tap 138 kV Rebuild	EES		J1100, J1159	5,700,000

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Indianola - Inverness 115 kV Rebuild	EES	J1007, J1020, J1049, J1060, J1073, J1078, J1091, J1093, J1095, J1097, J1100, J1113, J1125, J1130, J1137, J1146, J1155, J1156, J1157, J1163, J1168, J962, J964, J972		12,200,000
Isola - Belzoni 115 kV Terminal Equipment Upgrade	EES		J1070, J1113	500,000
J1023 POI - Osceola 161 kV Rebuild	EES		J1007, J1023, J1125, J1147, J1156, J972	500,000
J1113 POI - Delta 115 kV New Second Line	EES		J1113	18,400,000
J1125 POI - Wynne E. 161 kV Rebuild	EES		J1125, J1147, J972	5,000,000
J1125 POI - Wynne S. 161 kV Rebuild	EES	J1007, J1125, J1156		500,000
J1144 POI - Columbia 161 kV Uprate	CE	J1144		600,000
J1147 POI - Moses 161 kV New Second Line	EES		J1137, J1147	14,600,000
J1156 POI - Kuhn Road 161 kV Rebuild	EES		J1023, J1091, J1125, J1137, J1147, J1156	11,500,000
J1165 POI - Colonial Welsh Tap 138 kV Rebuild	EES		J1100, J1120, J1159, J1162, J1165, J1168, J1186	10,440,000
J1165 POI - Jennings 138 kV Rebuild	EES		J1120, J1159, J1162, J1165, J1168, J1186	29,640,000
J1165 POI - Lake Charles Bulk N. 138 kV Rebuild	EES		J1100, J1120, J1159, J1162, J1168, J1186	20,200,000
J1165 POI - Lowe Grout Road 115 kV Rebuild	EES		J1120, J1162, J1168, J1186	9,960,000
J1186 POI - Coley Creek 138 kV Rebuild	EES	J1186		7,500,000
J620 Tap - Macon Lake 115 kV Re-conductor	EES		J1093	5,000,000
J643 Tap - Marshall 138 kV Rebuild	EES		J1120, J1159, J1162, J1168, J1186	28,040,000
J683 Tap - Wilbert 138 kV Rebuild	EES		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186, J961	2,760,000
J830 Tap - Yandell Road 230 kV Rebuild	EES		J1113	12,376,000
J972 POI - Hughes 161 kV Rebuild	EES	J1147, J972		1,000,000
Jacinto - Pelican Road 138 kV Rebuild	EES	J1030, J1186		7,350,000
Jacksonville N. - Holland Bottoms 115 kV Rebuild	EES	J1097		11,700,000
Jacksonville N. - Jacksonville S. 115 kV Terminal Equipment Upgrade	EES	J1007, J1023, J1091, J1125, J1137, J1147, J1156		500,000
Jaguar - Esso Tap 230 kV Rebuild	EES		J961	6,640,000
Kuhn Road - Ebony 161 kV Rebuild	EES		J1023, J1125, J1137, J1147, J1156	3,500,000
Kuhn Road - West Memphis 161 kV New Line	EES	J1156	J1007, J1023, J1091, J1125, J1137, J1147, J1156	17,700,000
L673 Tap - Goosport 69 kV Rebuild	EES		J1165	2,468,000
LA Station 138/69 kV Replace Autotransformer	EES	J961		8,800,000
Lake Charles Bulk S. - Lowe Grout Road 138 kV Rebuild	EES		J1120, J1159, J1162, J1168, J1186	11,400,000
Marked Tree - Harrisburg 161 kV Rebuild	EES		J1007, J1023, J1060, J1125, J1147, J1156, J972	3,600,000
Missionary 161/69 kV Replace Autotransformer #1	CE	J1020		1,410,000
Missionary 161/69 kV Replace Autotransformer #2	CE	J1020		1,410,000
Moses - Forest City North 161 kV Rebuild	EES		J1147, J972	5,000,000

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Mt Olive 230 kV 40 MVar Capacitor Bank	EES	J1007, J1023, J1060, J1073, J1076, J1078, J1079, J1091, J1093, J1095, J1097, J1099, J1125, J1137, J1142, J1147, J1151, J1155, J1156, J1158, J1163, J1184, J961, J962, J972		2,500,000
Mt. Zion - Wyntx Tap 138 kV Rebuild	EES	J1159		7,620,000
Nelson - Penton Road 230 kV Rebuild	EES/ CLEC		J1100, J1120, J1159, J1162, J1165, J1168, J1186	23,800,000
New Long John - Dayton 138 kV Rebuild	EES		J1100, J1159, J1162, J1186	9,600,000
North Crowley - Scott 138 kV Rebuild	EES		J1165, J1166	28,200,000
Perryville - Baxter Wilson 500 kV Terminal Equipment Upgrade	EES		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	3,000,000
Ponderosa - Conroe 138 kV Rebuild	EES		J1100, J1159	5,100,000
Raywood - Pintail 138 kV Rebuild	EES	J1168		6,500,000
Richard - Colonial Academy 138 kV Rebuild	EES		J1165, J1166	17,960,000
Richard - Wells 500 kV Terminal Equipment Upgrade	EES		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	3,000,000
Richard 500/138 kV Second Autotransformer	EES		J1100, J1120, J1159, J1162, J1168, J1186	20,000,000
Rivtrin - Staley 138 kV Rebuild	EES		J1100, J1159	13,000,000
Rork - Penton Road 230 kV New Line	CLEC		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	175,000,000
Rork - Sellers Rd 230 kV Upgrade	CLEC		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	31,000,000
Sabine - Mud Lake 230 kV Rebuild	EES		J1100, J1120, J1159, J1162, J1168, J1186	81,500,000
SE Greenville - Greenville 115 kV Rebuild	EES		J1070	4,780,000
Tap Lake Charles Bulk S. - Jennings 138 kV into J1165 POI	EES		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,500,000
Tezcuco - Frisco 230 kV #1 & #2 Re-conductor ¹	EES		J1076, J1079, J1142, J1158	2,500,000
Twinkletown - Freeport 230 kV Rebuild	EES		J1070, J1093, J1095, J1113, J1163, J962	14,312,000
West Memphis 500/161 kV Second Autotransformer	EES		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	15,000,000
Wynne E. - Parkin 161 kV Rebuild	EES		J1125, J1147, J972	15,300,000

¹ This Network Upgrade is an Appendix B project in MTEP19 (Project ID 15605), therefore, as stated in section 6.1.1.1 of [BPM 015](#) if this project is approved by the Board within (1) calendar year of the GIA execution or execution of an amendment thereof, then the IC's will not be responsible for the cost of the project.

Table 4: Shared Network Upgrades (Planning level cost estimates)

Shared Network Upgrade	TO	Higher queued projects associated with SNU	Study projects associated with SNU	Cost of solution (\$)
Nelson – Goosport – Lake Charles Bulk N. 138 kV Rebuild	EES	J484	J1165	11,400,000
Kuhn Road – Ebony 161 kV Rebuild	EES	J934	J1007, J1060, J1125, J1156	4,900,000

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3) Analysis performed shows there are five projects for Shared Network Upgrade cost allocation.
- 4) A detailed summary of the estimated cost of Network Upgrades on a per project basis can be found in Appendix B – Network Upgrade Table Per Project (CEII).

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, states that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to all wind and solar projects included in the DPP 2018 April South study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter-based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meet the requirement to maintain 0.95 leading power factor, however, one project does not meet the requirement to provide reactive power capability corresponding to 0.95 lagging power factor, as highlighted in red below in Table 5. Additional reactive support will be needed for this project to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 5: FERC Order 827 Review Results

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement ?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1007	200	±65.737	2 x 14 MVar Cap	197.6	93.2	0.9044	196.8	110.6	-0.8718	Yes	
J1020	250	±82.171	2 x 17 MVar Cap	247.0	102.9	0.9231	245.3	149.0	-0.8547	Yes	
J1023	82	±61.5	N/A	80.9	44.7	0.8753	79.5	99.2	-0.6254	Yes	
J1030	153.5	±50.453	2 x 16 MVar Cap	150.6	56.1	0.9371	147.3	124.0	-0.7650	Yes	
J1049	250	±82.171	2 x 17 MVar Cap	246.7	95.9	0.9321	245.2	149.6	-0.8537	Yes	
J1060	500	±242.161	2 x 6 MVar Cap	488.9	210.0	0.9188	485.8	287.5	-0.8606	Yes	
J1070	100	±48.432	N/A	98.8	39.1	0.9298	98.2	64.3	-0.8366	Yes	
J1073	300	±145.297	N/A	295.8	95.3	0.9518	293.5	229.3	-0.7880	No	1.9
J1076	200	±96.864	N/A	197.3	66.7	0.9473	195.8	147.7	-0.7983	Yes	
J1078	300	±145.297	N/A	296.1	100.5	0.9469	294.5	214.8	-0.8079	Yes	
J1079	200	±96.864	N/A	197.3	67.8	0.9457	196.2	142.5	-0.8091	Yes	
J1091	150	±72.648	N/A	148.1	52.5	0.9425	147.1	106.7	-0.8095	Yes	
J1093	200	±96.864	N/A	197.5	70.8	0.9413	194.2	170.0	-0.7524	Yes	
J1095	300	±145.297	N/A	296.2	104.4	0.9431	294.1	216.9	-0.8048	Yes	
J1097	500	±242.161	N/A	493.7	172.8	0.9439	490.9	354.2	-0.8109	Yes	
J1099	200	±96.864	N/A	197.3	66.9	0.9470	196.0	145.5	-0.8029	Yes	
J1100	200	±96.864	N/A	197.2	66.0	0.9483	195.8	148.7	-0.7964	Yes	
J1113	85	±27.938	2 x 7 MVar Cap	84.1	30.1	0.9415	83.9	44.1	-0.8852	Yes	
J1120	145.0	±47.659	2 x 10 MVar Cap	142.2	48.7	0.9461	140.6	90.5	-0.8409	Yes	
J1125	135	±101.25	N/A	133.1	73.0	0.8768	130.3	171.3	-0.6054	Yes	
J1137	100	±75	N/A	97.8	58.8	0.8570	96.8	99.6	-0.6970	Yes	
J1142	20	±11.222	2 x 4.8 MVar Cap	19.6	17.7	0.7422	19.4	18.7	-0.7200	Yes	
J1144	96	±72	1 x 5 MVar Cap	91.9	61.4	0.8315	88.4	102.3	-0.6538	Yes	
J1147	100	±75	N/A	97.7	57.6	0.8614	96.0	106.3	-0.6702	Yes	
J1155	150	±72.6	2 x 24 MVar Cap	147.6	102.0	0.8227	147.1	87.1	-0.8605	Yes	
J1156	150	±72.6	2 x 24 MVar Cap	147.5	96.7	0.8363	145.7	126.4	-0.7554	Yes	
J1157	100	±48.4	2 x 15 MVar Cap	98.4	61.4	0.8484	97.3	84.2	-0.7562	Yes	
J1158	70	±33.9	2 x 11 MVar Cap	68.9	43.8	0.8439	68.2	56.7	-0.7690	Yes	
J1159	100	±48.4	2 x 11 MVar Cap	98.5	55.0	0.8731	97.3	82.7	-0.7620	Yes	
J1162	80	±38.7	2 x 12 MVar Cap	78.7	49.8	0.8450	78.0	64.1	-0.7726	Yes	

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement ?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1163	120	±58.1	2 x 13 MVar Cap	118.2	68.6	0.8649	117.0	93.4	-0.7815	Yes	
J1165	150	±72.6	2 x 23 MVar Cap	147.6	97.9	0.8334	145.5	127.2	-0.7529	Yes	
J1166	50	±24.2	2 x 7 MVar Cap	49.2	29.6	0.8569	48.7	40.6	-0.7681	Yes	
J1168	80.0	±38.7	2 x 12 MVar Cap	78.7	51.5	0.8368	78.3	63.5	-0.7767	Yes	
J1184	50	±24.527	2 x 7 MVar Cap	49.3	30.3	0.8520	48.9	41.5	-0.7624	Yes	
J1186	100	±48.4	2 x 14 MVar Cap	98.4	60.8	0.8507	97.5	79.9	-0.7735	Yes	
J961	150	±49.303	3 x 11 MVar Cap	148.4	60.1	0.9269	147.8	85.7	-0.8651	Yes	
J962	100	±32.868	2 x 13.5 MVar Cap	98.9	45.1	0.9099	98.6	38.5	-0.9315	Yes	
J964	70	±33.903	1 x 15 MVar Cap	68.8	37.4	0.8786	68.3	38.8	-0.8695	Yes	
J972	140	±67.805	2 x 4 MVar Cap	138.3	53.4	0.9329	137.1	108.8	-0.7833	Yes	
J988	200	±65.737	2 x 14 MVar Cap	197.4	93.8	0.9032	196.5	116.0	-0.8611	Yes	

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2018 April South models is the MTEP18 models with the Bench Cases including all pre-queued projects and associated Network Upgrades, while the Study Cases contain all of the interconnection requests in DPP 2018 April South Phase I, in addition to all the facilities in the Bench Cases.

- Bench Cases
 - APR18-PhaseI-2023SUM-Bench
- Study Cases
 - APR18-PhaseI-2023SUM-Study

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2018 April South generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 33.11.0 and TARA version 1901.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting NRIS must mitigate constraints found by using the Deliverability algorithm to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Thermal Analysis

The thermal analysis results for 2023 Summer show generator projects J1007, J1020, J1023, J1030, J1073, J1091, J1097, J1100, J1125, J1130, J1137, J1144, J1147, J1156, J1157, J1159, J1168, J1186, J961, J964, and J972 causing constraints. Several thermal constraints are mitigated by MTEP Project 9823, Target Appendix A in MTEP19, therefore no mitigation is required and no study project is assigned any cost for the mitigation of said constraints dependent upon approval of this project by the MISO Board of Directors in December 2019. In addition, the ERS service of study projects that are dependent upon approval of this project will be conditional on this MTEP project going into service. The details pertaining to the thermal analysis can be found in Appendix C – MISO ERS Analysis (CEII).

4.1. J1007

Thermal analysis determined that this generator contributes to the following constraints:

1. J1125 POI – Wynne S. 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the J1125 POI to Wynne S. 161 kV line. The cost estimate is \$0.5 million, with the project allocated 23.9% of the cost.

2. Indianola – Inverness 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Indianola to Inverness 115 kV line. The cost estimate is \$12.2 million, with the project allocated 0.9% of the cost.

3. Jacksonville N. – Jacksonville S. 115 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade terminal equipment on the Jacksonville N. to Jacksonville S. 115 kV line. The cost estimate is \$0.5 million, with the project allocated 28.1% of the cost.

4.2. J1020

Thermal analysis determined that this generator contributes to the following constraints:

1. Missionary 161/69 kV autotransformer #1

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to replace the Missionary 161/69 kV autotransformer #1. The cost estimate is \$1.41 million, with the project allocated 100% of the cost.

2. Missionary 161/69 kV autotransformer #2

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to replace the Missionary 161/69 kV autotransformer #2. The cost estimate is \$1.41 million, with the project allocated 100% of the cost.

3. Indianola – Inverness 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Indianola to Inverness 115 kV line. The cost estimate is \$12.2 million, with the project allocated 0.5% of the cost.

4.3. J1023

Thermal analysis determined that this generator contributes to the following constraints:

1. Jacksonville N. – Jacksonville S. 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade terminal equipment on the Jacksonville N. to Jacksonville S. 115 kV line. The cost estimate is \$0.5 million, with the project allocated 4.5% of the cost.

4.4. J1030

Thermal analysis determined that this generator contributes to the following constraints:

1. Jacinto – Pelican Road 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Jacinto to Pelican Road 138 kV line. The cost estimate is \$7.35 million, with the project allocated 39.2% of the cost.

4.5. J1073

Thermal analysis determined that this generator contributes to the following constraints:

1. Webre – Bayou Labutte 500 kV
2. Bayou Labutte – Willow Glen 500 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the two listed constraints. The planned upgrade is to build a new Big Cajun to Coly 500 kV line. The cost estimate is \$275 million, with the project allocated 29.2% of the cost.

3. McKnight – Coly 500 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 4.2% of the cost.

4. Indianola – Inverness 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Indianola to Inverness 115 kV line. The cost estimate is \$12.2 million, with the project allocated 4.2% of the cost.

4.6. J1091

Thermal analysis determined that this generator contributes to the following constraints:

1. Jacksonville N. – Jacksonville S. 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade terminal equipment on the Jacksonville N. to Jacksonville S. 115 kV line. The cost estimate is \$0.5 million, with the project allocated 4.0% of the cost.

4.7. J1097

Thermal analysis determined that this generator contributes to the following constraints:

1. Jacksonville N. – Holland Bottoms 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Jacksonville N. to Holland Bottoms 115 kV line. The cost estimate is \$11.7 million, with the project allocated 100% of the cost.

2. Indianola – Inverness 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Indianola to Inverness 115 kV line. The cost estimate is \$12.2 million, with the project allocated 0.5% of the cost.

4.8. J1100

Thermal analysis determined that this generator contributes to the following constraints:

1. Grimes 345/138 autotransformer

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 3.6% of the cost.

2. Indianola – Inverness 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Indianola to Inverness 115 kV line. The cost estimate is \$12.2 million, with the project allocated 3.6% of the cost.

4.9. J1125

Thermal analysis determined that this generator contributes to the following constraints:

1. J1125 POI – Wynne S. 161 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the J1125 POI to Wynne S. 161 kV line. The cost estimate is \$0.5 million, with the project allocated 47.1% of the cost.

2. Indianola – Inverness 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Indianola to Inverness 115 kV line. The cost estimate is \$12.2 million, with the project allocated 0.2% of the cost.

3. Jacksonville N. – Jacksonville S. 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade terminal equipment on the Jacksonville N. to Jacksonville S. 115 kV line. The cost estimate is \$0.5 million, with the project allocated 13.2% of the cost.

4.10. J1130

Thermal analysis determined that this generator contributes to the following constraints:

1. Franklin – Ray Braswell 500 kV
2. Franklin – Grand Gulf 500 kV
3. Franklin – McKnight 500 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the three listed constraints. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 39.1% of the cost.

4. Indianola – Inverness 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Indianola to Inverness 115 kV line. The cost estimate is \$12.2 million, with the project allocated 39.1% of the cost.

4.11. J1137

Thermal analysis determined that this generator contributes to the following constraints:

1. Jacksonville N. – Jacksonville S. 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade terminal equipment on the Jacksonville N. to Jacksonville S. 115 kV line. The cost estimate is \$0.5 million, with the project allocated 2.0% of the cost.

4.12. J1144

Thermal analysis determined that this generator contributes to the following constraints:

1. J1144 POI – Columbia 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1144 POI to Columbia 161 kV line. The cost estimate is \$0.6 million, with the project allocated 100% of the cost.

4.13. J1147

Thermal analysis determined that this generator contributes to the following constraints:

1. J972 POI – Hughes 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the J972 POI to Hughes 161 kV line. The cost estimate is \$1.0 million, with the project allocated 21.3% of the cost.

2. Jacksonville N. – Jacksonville S. 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade terminal equipment on the Jacksonville N. to Jacksonville S. 115 kV line. The cost estimate is \$0.5 million, with the project allocated 1.7% of the cost.

4.14. J1156

Thermal analysis determined that this generator contributes to the following constraints:

1. J1125 POI – Wynne S. 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the J1125 POI to Wynne S. 161 kV line. The cost estimate is \$0.5 million, with the project allocated 28.9% of the cost.

2. J1156 POI – Parkin 161 kV
3. Kuhn Road – Ebony 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the two listed constraints. The planned upgrade is to build a new Kuhn Road to West Memphis 161 kV line. The cost estimate is \$17.7 million, with the project allocated 72.3% of the cost.

4. Jacksonville N. – Jacksonville S. 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade terminal equipment on the Jacksonville N. to Jacksonville S. 115 kV line. The cost estimate is \$0.5 million, with the project allocated 46.6% of the cost.

4.15. J1157

Thermal analysis determined that this generator contributes to the following constraints:

1. Franklin – Ray Braswell 500 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 6.6% of the cost.

2. Indianola – Inverness 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Indianola to Inverness 115 kV line. The cost estimate is \$12.2 million, with the project allocated 6.6% of the cost.

4.16. J1159

Thermal analysis determined that this generator contributes to the following constraints:

1. Mt. Zion – Wyntx Tap 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Mt. Zion to Wyntx Tap 138 kV line. The cost estimate is \$7.62 million, with the project allocated 100% of the cost.

4.17. J1168

Thermal analysis determined that this generator contributes to the following constraints:

1. Raywood – Pintail 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Raywood to Pintail 138 kV line. The cost estimate is \$6.5 million, with the project allocated 100% of the cost.

2. Indianola – Inverness 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Indianola to Inverness 115 kV line. The cost estimate is \$12.2 million, with the project allocated 2.0% of the cost.

4.18. J1186

Thermal analysis determined that this generator contributes to the following constraints:

1. Jacinto – Pelican Road 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Jacinto to Pelican Road 138 kV line. The cost estimate is \$7.35 million, with the project allocated 60.8% of the cost.

2. J1186 POI – Coley Creek 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the J1186 POI to Coley Creek 138 kV line. The cost estimate is \$7.5 million, with the project allocated 100% of the cost.

3. Coley Creek – Shepherd 138 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Coley Creek to Shepherd 138 kV line. The cost estimate is \$4.52 million, with the project allocated 100% of the cost.

4.19. J961

Thermal analysis determined that this generator contributes to the following constraints:

1. LA Station 138/69 kV autotransformer

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the replacement of the LA Station 138/69 kV autotransformer. The cost estimate is \$8.8 million, with the project allocated 100% of the cost.



4.20. J964

Thermal analysis determined that this generator contributes to the following constraints:

1. Franklin – Ray Braswell 500 kV
2. Franklin – McKnight 500 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the two listed constraints. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 5.2% of the cost.

3. Indianola – Inverness 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Indianola to Inverness 115 kV line. The cost estimate is \$12.2 million, with the project allocated 5.2% of the cost.

4.21. J972

Thermal analysis determined that this generator contributes to the following constraints:

1. J972 POI – Hughes 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the J972 POI to Hughes 161 kV line. The cost estimate is \$1.0 million, with the project allocated 78.7% of the cost.

2. Indianola – Inverness 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to rebuild the Indianola to Inverness 115 kV line. The cost estimate is \$12.2 million, with the project allocated 0.6% of the cost.

5. Voltage Analysis

The voltage analysis results for 2023 Summer show generator projects J1007, J1020, J1023, J1049, J1060, J1073, J1076, J1078, J1079, J1091, J1093, J1095, J1097, J1099, J1113, J1125, J1137, J1142, J1146, J1147, J1151, J1155, J1156, J1157, J1158, J1163, J1168, J1184, J961, J962, J964, and J972 causing voltage violations. Several voltage constraints are mitigated by MTEP Project 9823, Target Appendix A in MTEP19, therefore no mitigation is required and no study project is assigned any cost for the mitigation of said constraints. The details pertaining to the voltage analysis can be found in Appendix C – MISO ERIIS Analysis (CEII).

5.1. J1007

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 0.9% of the cost.

2. Worst Voltage at Grambling 230 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 1.0% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 1.0% of the cost.

5.2. J1020

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 0.5% of the cost.

5.3. J1023

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.1% of the cost.

2. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated .1% of the cost.

5.4. J1049

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 15.5% of the cost.

5.5. J1060

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 1.0% of the cost.

2. Worst Voltage at Grambling 230 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 2.2% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 2.4% of the cost.

5.6. J1073

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 4.2% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 12.5% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 12.4% of the cost.

5.7. J1076

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 7.3% of the cost.

2. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 7.1% of the cost.

5.8. J1078

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 1.1% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 1.0% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 1.1% of the cost.

5.9. J1079

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 16.0% of the cost.

2. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 16.0% of the cost.

5.10. J1091

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 0.3% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.7% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.8% of the cost.

5.11. J1093

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 6.5% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.4% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.8% of the cost.

5.12. J1095

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 2.4% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.8% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.9% of the cost.

5.13. J1097

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 0.5% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 5.3% of the cost.

3. Worst Voltage at Mt. Olive 230 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 5.6% of the cost.

5.14. J1099

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 15.1% of the cost.

2. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 15.0% of the cost.

5.15. J1113

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 4.3% of the cost.

5.16. J1125

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 0.2% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.4% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.5% of the cost.

5.17. J1137

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 0.7% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVar capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.8% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVar capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.8% of the cost.

5.18. J1142

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVar capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.5% of the cost.

2. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVar capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.5% of the cost.

5.19. J1146

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 0.04% of the cost.

5.20. J1147

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVar capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.3% of the cost.

2. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVar capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.3% of the cost.

5.21. J1151

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 25.3% of the cost.

2. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 24.8% of the cost.

5.22. J1155

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 0.5% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.4% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.5% of the cost.

5.23. J1156

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 0.9% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.8% of the cost.

3. Worst Voltage at Mt. Olive 230 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.8% of the cost.

5.24. J1157

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 6.6% of the cost.

5.25. J1158

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 2.2% of the cost.

2. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 2.0% of the cost.

5.26. J1163

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 2.4% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.3% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.3% of the cost.

5.27. J1168

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 2.0% of the cost.

5.28. J1184

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 2.6% of the cost.

2. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 2.4% of the cost.

5.29. J961

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 3.3% of the cost.

2. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 3.0% of the cost.

5.30. J962

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 1.0% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVAR capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.3% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.3% of the cost.



5.31. J964

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 5.2% of the cost.

5.32. J972

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to build new Baxter Wilson to Franklin and Franklin to Big Cajun 500 kV lines. The cost estimate is \$752.5 million, with the project allocated 0.6% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a two 20 MVar capacitor banks at the Danville 115 kV substation. The cost estimate is \$4.5 million, with the project allocated 0.6% of the cost.

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 40 MVar capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.6% of the cost.

6. Stability Analysis

Stability analysis was not included as part of the DPP 2018 April South Phase I analysis. Stability analysis will be completed as part of the Phase II and Phase III analysis.

7. Short Circuit Analysis

Short circuit analysis was not included as part of the DPP 2018 April South Phase I analysis. Short circuit analysis will be completed as part of the Phase II analysis.

8. Affected System Impact Study

Affected System Impact Studies were not included as part of the DPP 2018 April South Phase I analysis. Affected System Impact Studies will be completed as part of the Phase II and Phase III analysis.

9. Deliverability Analysis

9.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource project or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The



wind and solar generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TENT).

The MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

9.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator's output is not deliverable.

9.3. Deliverability Study Results

The limiting constraints (monitored facilities – contingency facility pairs) seen in the deliverability analysis for the 2023 Summer case are summarized in Appendix D – MISO Deliverability Analysis (CEII).

9.3.1. J1007

This generator is determined to be deliverable up to 91.5 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 6 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 6: NRIS Results for J1007

J1007 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)				91.5 MW (45.76%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIIS Network Upgrade	Projects Associated with ERIIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Perryville - Baxter Wilson 500 kV	117.3	5.86%	No		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	202,022	3,000,000
West Memphis 500/161 kV autotransformer	144.1	45.86%	No		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	3,600,940	155,000,000
Ebony - West Memphis Lehi-W 161 kV	155.0	45.65%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
West Memphis Lehi-W - West Memphis 161 kV	158.8	45.65%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Ebony - West Memphis Lehi-E 161 kV	174.5	43.21%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
West Memphis Lehi-E - West Memphis Polk Ave 161 kV	175.1	43.21%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Marked Tree - Harrisburg 161 kV	184.2	42.63%	No		J1007, J1023, J1060, J1125, J1147, J1156, J972	1,125,179	3,600,000
J1023 POI - Osceola 161 kV	198.3	15.21%	No		J1007, J1023, J1125, J1147, J1156, J972	87,909	500,000
West Memphis Polk Ave - West Memphis 161 kV	200.0	43.21%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000

9.3.2. J1020

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 7 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 7: NRIS Results for J1020

J1020 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	73.4	8.33%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	3,912,080	275,000,000
Big Cajun - Webre 500 kV	202.3	8.81%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	4,137,506	275,000,000
Coly - Willow Glen 500 kV	250.0	8.14%	No		J1020, J1049, J1070, J1113, J1130, J1144, J1157, J1184, J964, J988	268,805	3,000,000

9.3.3. J1023

This generator is determined to be deliverable up to 37.5 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 8 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 8: NRIS Results for J1023

J1023 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				37.5 MW (45.76%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Perryville - Baxter Wilson 500 kV	48.1	5.11%	No		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	72,408	3,000,000
West Memphis 500/161 kV autotransformer	58.3	16.70%	No		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	539,380	15,000,000
Kuhn Road - Ebony 161 kV	59.1	11.80%	Yes	J1156	J1023, J1125, J1137, J1147, J1156	171,963	21,200,000
Ebony - West Memphis Lehi-W 161 kV	63.6	10.84%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
West Memphis Lehi-W - West Memphis 161 kV	63.7	10.84%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Parkin - Twist 161 kV	65.1	10.87%	Yes	J1156	J1023, J1125, J1147, J1156	0	17,700,000
Ebony - West Memphis Lehi-E 161 kV	65.2	10.26%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Marked Tree - Twist 161 kV	69.1	10.87%	Yes	J1156	J1023, J1125, J1147, J1156	0	17,700,000
J1156 POI - Kuhn Road 161 kV	71.5	12.96%	No		J1023, J1091, J1125, J1137, J1147, J1156	766,255	11,500,000

West Memphis Lehi-E - West Memphis Polk Ave 161 kV	71.8	10.26%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Marked Tree - Harrisburg 161 kV	75.5	8.90%	No		J1007, J1023, J1060, J1125, J1147, J1156, J972	96,312	3,600,000
J1023 POI - Osceola 161 kV	81.3	73.90%	No		J1007, J1023, J1125, J1147, J1156, J972	175,118	500,000
West Memphis Polk Ave - West Memphis 161 kV	82.0	10.26%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000

9.3.4. J1049

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 9 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 9: NRIS Results for J1049

J1049 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	73.4	7.21%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	3,386,086	275,000,000
Big Cajun - Webre 500 kV	202.3	8.22%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	3,860,420	275,000,000
Coly - Willow Glen 500 kV	250.0	8.06%	No		J1020, J1049, J1070, J1113, J1130, J1144, J1157, J1184, J964, J988	266,163	3,000,000

9.3.5. J1060

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 10 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 10: NRIS Results for J1060

J1060 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Perryville - Baxter Wilson 500 kV	444.1	5.18%	No		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	560,348	3,000,000
Marked Tree - Harrisburg 161 kV	500.0	8.31%	No		J1007, J1023, J1060, J1125, J1147, J1156, J972	548,337	3,600,000

9.3.6. J1070

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 11 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 11: NRIS Results for J1070

J1070 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
SE Greenville - Greenville 115 kV	0.0	11.79%	No		J1070	4,780,000	4,780,000
McKnight - Coly 500 kV	22.1	9.78%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	1,837,222	275,000,000
Isola - Belzoni 115 kV	29.4	5.52%	No		J1070, J1113	279,579	500,000
Big Cajun - Webre 500 kV	42.9	9.93%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	1,865,400	275,000,000
Greenville - Arcola 115 kV	80.9	11.79%	No		J1070	13,096,000	13,096,000
Coly - Willow Glen 500 kV	93.9	7.81%	No		J1020, J1049, J1070, J1113, J1130, J1144, J1157, J1184, J964, J988	103,163	3,000,000
Twinkletown - Freeport 230 kV	97.4	6.57%	No		J1070, J1093, J1095, J1113, J1163, J962	247,480	14,312,000
Arcola - Hollandale 115 kV	100.0	11.93%	No		J1070	12,886,000	12,886,000

9.3.7. J1073

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 12 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 12: NRIS Results for J1073

J1073 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	0.0	6.99%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	3,939,319	275,000,000
Bayou Labutte 500/230 kV autotransformer	0.0	15.14%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	7,665,356	18,000,000
Webre - Bayou Labutte 500 kV	78.0	22.22%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	12,522,414	275,000,000
Bayou Labutte - Willow Glen 500 kV	220.9	19.15%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	10,792,270	275,000,000
Bayou Labutte - Iberville 230 kV	221.6	15.14%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	6,685,893	15,700,000
Big Cajun - Fancy Point 500 kV	300.0	38.20%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	21,528,183	275,000,000

9.3.8. J1076

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 13 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 13: NRIS Results for J1076

J1076 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcucu - Frisco 230 kV #1	0.0	12.42%	No		J1076, J1079, J1142, J1158	424,646	2,500,000
Tezcucu - Frisco 230 kV #2	200.0	12.36%	No		J1076, J1079, J1142, J1158	422,414	2,500,000

9.3.9. J1078

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 14 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 14: NRIS Results for J1078

J1078 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIS Network Upgrade	Projects Associated with ERIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	176.0	5.08%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	2,862,910	275,000,000
West Memphis 500/161 kV autotransformer	300.0	8.21%	No		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	965,503	15,000,000

9.3.10. J1079

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 15 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 15: NRIS Results for J1079

J1079 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIS Network Upgrade	Projects Associated with ERIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcucu - Frisco 230 kV #1	0.0	18.65%	No		J1076, J1079, J1142, J1158	637,380	2,500,000
Tezcucu - Frisco 230 kV #2	200.0	18.57%	No		J1076, J1079, J1142, J1158	634,646	2,500,000

9.3.11. J1091

This generator is determined to be deliverable up to 68.6 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 16 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 16: NRIS Results for J1091

J1091 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)				68.6 MW (45.76%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIIS Network Upgrade	Projects Associated with ERIIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Perryville - Baxter Wilson 500 kV	88.0	6.15%	No		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	158,201	3,000,000
West Memphis 500/161 kV autotransformer	108.0	13.28%	No		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	784,986	15,000,000
Ebony - West Memphis Lehi-W 161 kV	116.3	8.65%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
West Memphis Lehi-W - West Memphis 161 kV	119.1	8.65%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Ebony - West Memphis Lehi-E 161 kV	126.4	8.19%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
J1156 POI - Kuhn Road 161 kV	130.9	5.36%	No		J1023, J1091, J1125, J1137, J1147, J1156	579,709	11,500,000
West Memphis Lehi-E - West Memphis Polk Ave 161 kV	148.7	8.19%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
West Memphis Polk Ave - West Memphis 161 kV	150.0	8.19%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000

9.3.12. J1093

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 17 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 17: NRIS Results for J1093

J1093 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIIS Network Upgrade	Projects Associated with ERIIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Perryville - Baxter Wilson 500 kV	0.0	5.99%	No		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	260,120	3,000,000
McKnight - Coby 500 kV	58.7	6.40%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	2,404,544	275,000,000
Big Cajun - Webre 500 kV	187.7	6.19%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	2,325,645	275,000,000
Twinkletown - Freeport 230 kV	199.7	5.49%	No		J1070, J1093, J1095, J1113, J1163, J962	413,596	14,312,000
J620 Tap - Macon Lake 115 kV	200.0	22.63%	No		J1093	5,000,000	5,000,000

9.3.13. J1095

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 18 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 18: NRIS Results for J1095

J1095 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIIS Network Upgrade	Projects Associated with ERIIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	88.1	5.35%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	3,015,073	275,000,000
Big Cajun - Webre 500 kV	281.6	5.00%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	2,817,825	275,000,000
Twinkletown - Freeport 230 kV	300.0	77.79%	No		J1070, J1093, J1095, J1113, J1163, J962	8,790,608	14,312,000

9.3.14. J1097

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 19 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 19: NRIS Results for J1097

J1097 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIIS Network Upgrade	Projects Associated with ERIIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Perryville - Baxter Wilson 500 kV	500.0	10.55%	No		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	1,081,084	3,000,000

9.3.15. J1099

This generator is determined to be deliverable up to 198.2 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 20 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 20: NRIS Results for J1099

J1099 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		198.2 MW (99.12%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIS Network Upgrade	Projects Associated with ERIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Addis - Tiger 230 kV	200.0	33.74%	No		J1099	24,560,000	24,560,000

9.3.16. J1100

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 21 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 21: NRIS Results for J1100

J1100 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIS Network Upgrade	Projects Associated with ERIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J1165 POI - Lake Charles Bulk N. 138 kV	0.0	5.19%	No		J1100, J1120, J1159, J1162, J1168, J1186	6,295,974	195,200,000
Nelson - Penton Road 230 kV	0.0	10.58%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	4,764,819	23,800,000
Cocodrie - Ville Platte 230 kV	0.0	6.24%	No		J1100, J1120, J1159, J1162, J1168, J1186	1,857,344	175,000,000
J1165 POI - Colonial Welsh Tap 138 kV	0.0	5.19%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	2,671,154	185,440,000
Bayou Labutte 500/230 kV autotransformer	0.0	5.63%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,900,306	18,000,000
Sabine - Mud Lake 230 kV	0.0	13.66%	No		J1100, J1120, J1159, J1162, J1168, J1186	19,033,608	81,500,000
Webre - Bayou Labutte 500 kV	0.0	8.02%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	3,013,194	275,000,000
Richard 500/138 kV autotransformer	0.0	8.21%	No		J1120, J1162, J1168, J1186, J1159, J1100	4,705,781	20,000,000
Ville Platte - West Fork 230 kV	4.5	6.24%	No		J1100, J1120, J1159, J1162, J1168, J1186	1,857,344	175,000,000
Rork - Sellers Rd 230 kV	22.4	9.74%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	5,778,424	31,000,000
Henning - Hebert 138 kV	52.0	5.19%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,706,641	176,500,000
Bayou Labutte - Willow Glen 500 kV	58.0	6.93%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,603,670	275,000,000
Hebert - Bayou Cove 138 kV	60.7	5.19%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,544,810	175,000,000
Colonial Welsh Tap - Henning 138 kV	78.5	5.19%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,544,810	175,000,000

New Long John - Dayton 138 kV	90.9	5.22%	No		J1100, J1159, J1162, J1186	2,399,885	9,600,000
J1166 POI - Richard 138 kV	95.5	5.25%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,481,844	57,200,000
J1166 POI - Bayou Cove 138 kV	112.6	5.25%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	2,481,844	57,200,000
J683 Tap - Wilbert 138 kV	122.5	5.44%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186, J961	167,236	2,760,000
Grimes 345/138 kV autotransformer #1	124.7	23.36%	No		J1100	7,520,927	15,000,000
Grimes 345/138 kV autotransformer #2	129.0	23.23%	No		J1100	7,479,073	15,000,000
Huntsville - Wyntx Tap 138 kV	130.3	10.62%	No		J1100, J1159	1,805,161	5,700,000
Nelson 500/230 kV autotransformer	137.0	21.52%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	6,405,456	175,000,000
Champagne - Bobcat 138 kV	137.8	5.44%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,186,031	6,728,000
Rivtrin - Staley 138 kV	145.7	8.21%	No		J1100, J1159	6,531,824	13,000,000
Ponderosa - Conroe 138 kV	147.2	16.88%	No		J1100, J1159	3,811,733	5,100,000
Bayou Labutte - Iberville 230 kV	147.8	5.63%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,657,489	15,700,000
Big Cajun - Fancy Point 500 kV	160.6	14.75%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	5,541,722	275,000,000
Grimes - Mt. Zion 138 kV	175.3	10.62%	No		J1100	27,125,000	27,125,000
Nelson - Richard 500 kV	177.8	27.93%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	8,313,401	175,000,000
Richard - Wells 500 kV	200.0	25.70%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	544,709	3,000,000

9.3.17. J1113

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 22 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 22: NRIS Results for J1113

J1113 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIIS Network Upgrade	Projects Associated with ERIIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Canton - Pickens 230 kV	0.0	5.50%	No		J1113	37,160,000	37,160,000
J830 Tap - Yandell Road 230 kV	0.0	5.50%	No		J1113	12,376,000	12,376,000
McKnight - Coly 500 kV	18.8	8.76%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	1,398,768	275,000,000
Isola - Belzoni 115 kV	25.0	5.12%	No		J1070, J1113	220,421	500,000
Big Cajun - Webre 500 kV	68.8	8.79%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	1,403,559	275,000,000
Coly - Willow Glen 500 kV	75.3	6.95%	No		J1020, J1049, J1070, J1113, J1130, J1144, J1157, J1184, J964, J988	78,033	3,000,000
J1113 POI - Delta 115 kV	77.3	100.00%	No		J1113	9,200,000	18,400,000
Ruleville - Schlater 115 kV	79.8	100.00%	No		J1113	9,200,000	18,400,000
Twinkletown - Freeport 230 kV	85.0	11.76%	No		J1070, J1093, J1095, J1113, J1163, J962	376,530	14,312,000

9.3.18. J1120

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 23 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 23: NRIS Results for J1120

J1120 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIIS Network Upgrade	Projects Associated with ERIIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Nelson - Goosport 138 kV	0.0	5.53%	No		J1120, J1162, J1168	1,193,359	175,000,000
Nelson - Lake Charles Bulk S. 138 kV	0.0	5.55%	No		J1120, J1162, J1168	1,197,674	175,000,000
Lake Charles Bulk N. - Goosport 138 kV	0.0	5.53%	No		J1120, J1162, J1168	1,193,359	175,000,000
Lake Charles Bulk S. - Lowe Grout Rd 138 kV	0.0	6.57%	No		J1120, J1159, J1162, J1168, J1186	4,955,258	186,400,000
J1165 POI - Lake Charles Bulk N. 138 kV	0.0	7.23%	No		J1100, J1120, J1159, J1162, J1168, J1186	6,358,752	195,200,000
Nelson - Penton Road 230 kV	0.0	15.50%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	5,060,931	23,800,000
Cocodrie - Ville Platte 230 kV	0.0	6.94%	No		J1100, J1120, J1159, J1162, J1168, J1186	1,497,633	175,000,000
J1165 POI - Colonial Welsh Tap 138 kV	0.0	7.23%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	2,697,788	185,440,000

J1165 POI - Lowe Grout Rd 138 kV	0.0	6.10%	No		J1120, J1162, J1168, J1186	5,049,411	184,960,000
J1165 POI - Jennings 138 kV	0.0	6.81%	No		J1120, J1159, J1162, J1165, J1168, J1186	6,235,775	204,640,000
Bayou Labutte 500/230 kV autotransformer	0.0	7.07%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,730,106	18,000,000
Sabine - Mud Lake 230 kV	0.0	18.75%	No		J1100, J1120, J1159, J1162, J1168, J1186	18,941,296	81,500,000
Webre - Bayou Labutte 500 kV	0.0	9.48%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,582,255	275,000,000
Richard 500/138 kV autotransformer	0.0	11.17%	No		J1120, J1162, J1168, J1186, J1159, J1100	4,641,729	20,000,000
Ville Platte - West Fork 230 kV	0.0	6.94%	No		J1100, J1120, J1159, J1162, J1168, J1186	1,497,633	175,000,000
Rork - Sellers Rd 230 kV	9.4	13.24%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	5,694,774	31,000,000
Bayou Cove - Richard 138 kV	16.3	6.49%	No		J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,224,323	57,200,000
Henning - Hebert 138 kV	37.7	7.23%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,723,658	176,500,000
Bayou Labutte - Willow Glen 500 kV	42.0	8.21%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,236,320	275,000,000
Hebert - Bayou Cove 138 kV	44.0	7.23%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,560,214	175,000,000
Colonial Welsh Tap - Henning 138 kV	65.9	7.23%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,560,214	175,000,000
J1166 POI - Richard 138 kV	69.2	7.31%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,505,362	57,200,000
J1166 POI - Bayou Cove 138 kV	81.6	7.31%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	2,505,362	57,200,000
J683 Tap - Wilbert 138 kV	94.5	6.96%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186, J961	155,124	2,760,000
Nelson 500/230 kV autotransformer	99.3	29.68%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	6,404,861	175,000,000
Champagne - Bobcat 138 kV	106.8	6.96%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,100,131	6,728,000
Bayou Labutte - Iberville 230 kV	107.1	7.07%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,509,036	15,700,000
Big Cajun - Fancy Point 500 kV	115.6	18.69%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	5,090,964	275,000,000
J643 Tap - Marshall 138 kV	127.1	8.14%	No		J1120, J1159, J1162, J1168, J1186	9,994,749	28,040,000
Nelson - Richard 500 kV	128.9	36.61%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	7,900,336	175,000,000
Richard - Wells 500 kV	145.0	33.49%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	533,292	3,000,000

9.3.19. J1125

This generator is determined to be deliverable up to 61.8 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 24 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 24: NRIS Results for J1125

J1125 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)		61.8 MW (45.76%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Perryville - Baxter Wilson 500 kV	79.2	6.12%	No		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	142,158	3,000,000
West Memphis 500/161 kV autotransformer	96.0	32.56%	No		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	1,726,793	15,000,000
Kuhn Road - Ebony 161 kV	97.2	18.89%	Yes	J1156	J1023, J1125, J1137, J1147, J1156	453,216	21,200,000
Ebony - West Memphis Lehi-W 161 kV	104.6	18.10%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
West Memphis Lehi-W - West Memphis 161 kV	104.8	18.10%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Parkin - Twist 161 kV	107.2	23.61%	Yes	J1156	J1023, J1125, J1147, J1156	0	17,700,000
Ebony - West Memphis Lehi-E 161 kV	107.4	17.13%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Marked Tree - Twist 161 kV	108.6	23.61%	Yes	J1156	J1023, J1125, J1147, J1156	0	17,700,000
J1125 POI - Wynne E. 161 kV	113.8	63.20%	No		J1125, J1147, J972	2,494,883	5,000,000
J1156 POI - Kuhn Road 161 kV	117.8	21.49%	No		J1023, J1091, J1125, J1137, J1147, J1156	2,091,821	11,500,000
West Memphis Lehi-E - West Memphis Polk Ave 161 kV	118.2	17.13%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Marked Tree - Harrisburg 161 kV	119.5	25.17%	No		J1007, J1023, J1060, J1125, J1147, J1156, J972	448,429	3,600,000
Hughes - West Memphis 161 kV	124.3	14.97%	No		J1125, J1147, J1156, J972	4,130,544	28,000,000
J1023 POI - Osceola 161 kV	129.2	16.59%	No		J1007, J1023, J1125, J1147, J1156, J972	64,722	500,000
Wynne E. - Parkin 161 kV	133.9	63.20%	No		J1125, J1147, J972	7,634,341	15,300,000
West Memphis Polk Ave - West Memphis 161 kV	135.0	17.13%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000

9.3.20. J1130

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 25 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 25: NRIS Results for J1130

J1130 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	325.8	6.29%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	13,310,278	275,000,000
Big Cajun - Webre 500 kV	133.9	15.03%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	13,834,957	275,000,000
Coly - Willow Glen 500 kV	1110.0	11.64%	No		J1020, J1049, J1070, J1113, J1130, J1144, J1157, J1184, J964, J988	1,671,716	9,000,000

9.3.21. J1137

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 26 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 26: NRIS Results for J1137

J1137 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Perryville - Baxter Wilson 500 kV	58.7	5.34%	No		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	118,837	3,000,000
West Memphis 500/161 kV autotransformer	71.1	10.44%	No		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	411,995	15,000,000
Kuhn Road - Ebony 161 kV	72.0	5.20%	Yes	J1156	J1023, J1125, J1137, J1147, J1156	92,415	21,200,000
Ebony - West Memphis Lehi-W 161 kV	72.9	5.31%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
J1147 POI - Moses 161 kV	77.5	5.32%	No		J1137, J1147	225,074	14,600,000
West Memphis Lehi-W - West Memphis 161 kV	79.4	5.31%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Ebony - West Memphis Lehi-E 161 kV	84.3	5.03%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
J1156 POI - Kuhn Road 161 kV	87.2	5.93%	No		J1023, J1091, J1125, J1137, J1147, J1156	427,572	11,500,000
West Memphis Lehi-E - West Memphis Polk Ave 161 kV	99.2	5.03%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
West Memphis Polk Ave - West Memphis 161 kV	100.0	5.03%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000

9.3.22. J1142

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 27 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 27: NRIS Results for J1142

J1142 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcuc - Frisco 230 kV #1	0.0	12.42%	No		J1076, J1079, J1142, J1158	42,446	2,500,000
Tezcuc - Frisco 230 kV #2	20.0	12.36%	No		J1076, J1079, J1142, J1158	42,241	2,500,000

9.3.23. J1144

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 28 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 28: NRIS Results for J1144

J1144 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	28.2	7.90%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	1,424,692	275,000,000
Big Cajun - Webre 500 kV	77.7	8.82%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	1,590,606	275,000,000
Coly - Willow Glen 500 kV	96.0	8.38%	No		J1020, J1049, J1070, J1113, J1130, J1144, J1157, J1184, J964, J988	106,264	3,000,000

9.3.24. J1146

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 29 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 29: NRIS Results for J1146

J1146 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIS Network Upgrade	Projects Associated with ERIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	5.9	5.08%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	95,430	275,000,000
West Memphis 500/161 kV autotransformer	10.0	8.21%	No		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	32,183	15,000,000

9.3.25. J1147

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 30 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 30: NRIS Results for J1147

J1147 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIS Network Upgrade	Projects Associated with ERIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Perryville - Baxter Wilson 500 kV	58.7	5.10%	No		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	110,254	3,000,000
West Memphis 500/161 kV autotransformer	59.2	26.65%	No		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	1,047,432	15,000,000
Shoffner - Bailey 161 kV	62.8	100.00%	No		J1147	3,867,242	14,600,000
Shoffner - Newport East 161 kV	71.1	100.00%	No		J1147	3,867,242	14,600,000
Kuhn Road - Ebony 161 kV	72.9	6.56%	Yes	J1156	J1023, J1125, J1137, J1147, J1156	116,585	21,200,000
Newport - Newport East 161 kV	72.9	100.00%	No		J1147	3,867,242	14,600,000
J1147 POI - Moses 161 kV	77.6	71.49%	No		J1137, J1147	2,773,199	14,600,000
Parkin - Twist 161 kV	79.5	9.72%	Yes	J1156	J1023, J1125, J1147, J1156	0	17,700,000
Marked Tree - Twist 161 kV	80.4	9.72%	Yes	J1156	J1023, J1125, J1147, J1156	0	17,700,000
J1125 POI - Wynne E. 161 kV	84.3	32.40%	No		J1125, J1147, J972	947,424	5,000,000
J1156 POI - Kuhn Road 161 kV	87.5	7.95%	No		J1023, J1091, J1125, J1137, J1147, J1156	573,220	11,500,000
Marked Tree - Harrisburg 161 kV	88.5	16.85%	No		J1007, J1023, J1060, J1125, J1147, J1156, J972	222,370	3,600,000
Hughes - West Memphis 161 kV	92.1	21.60%	No		J1125, J1147, J1156, J972	4,414,744	28,000,000
J1023 POI - Osceola 161 kV	94.6	10.51%	No		J1007, J1023, J1125, J1147, J1156, J972	30,372	500,000

Moses - Forest City North 161 kV	95.7	32.40%	No		J1147, J972	1,890,977	5,000,000
Wynne E. - Parkin 161 kV	100.0	32.40%	No		J1125, J1147, J972	2,899,117	15,300,000

9.3.26. J1151

This generator is found to be fully deliverable up to 219 MW without any Network Upgrades.

9.3.27. J1155

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 31 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 31: NRIS Results for J1155

J1155 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIIS Network Upgrade	Projects Associated with ERIIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	99.8	5.87%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	1,434,273	275,000,000
West Memphis 500/161 kV autotransformer	150.0	42.97%	No		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	482,752	15,000,000

9.3.28. J1156

This generator is determined to be deliverable up to 68.6 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 32 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 32: NRIS Results for J1156

J1156 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)				68.6 MW (45.76%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIIS Network Upgrade	Projects Associated with ERIIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Perryville - Baxter Wilson 500 kV	88.0	5.87%	No		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	151,764	3,000,000
West Memphis 500/161 kV autotransformer	106.7	46.72%	No		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	2,751,273	15,000,000
Kuhn Road - Ebony 161 kV	108.0	56.38%	Yes	J1156	J1023, J1125, J1137, J1147, J1156	2,665,820	21,200,000
Ebony - West Memphis Lehi- W 161 kV	116.3	45.39%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
West Memphis Lehi-W - West Memphis 161 kV	116.4	45.39%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Parkin - Twist 161 kV	119.1	34.28%	Yes	J1156	J1023, J1125, J1147, J1156	0	17,700,000

Ebony - West Memphis Lehi-E 161 kV	119.3	42.97%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Marked Tree - Twist 161 kV	126.4	34.28%	Yes	J1156	J1023, J1125, J1147, J1156	0	17,700,000
J1156 POI - Kuhn Road 161 kV	130.9	65.29%	No		J1023, J1091, J1125, J1137, J1147, J1156	7,061,423	11,500,000
West Memphis Lehi-E - West Memphis Polk Ave 161 kV	131.3	42.97%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000
Marked Tree - Harrisburg 161 kV	132.7	32.91%	No		J1007, J1023, J1060, J1125, J1147, J1156, J972	651,472	3,600,000
Hughes - West Memphis 161 kV	138.2	6.72%	No		J1125, J1147, J1156, J972	2,060,214	28,000,000
J1023 POI - Osceola 161 kV	148.0	18.41%	No		J1007, J1023, J1125, J1147, J1156, J972	84,267	500,000
J1156 POI - Parkin 161 kV	148.7	100.00%	Yes	J1156	J1156	0	17,700,000
West Memphis Polk Ave - West Memphis 161 kV	150.0	42.97%	Yes	J1156	J1007, J1023, J1091, J1125, J1137, J1156	0	17,700,000

9.3.29. J1157

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 33 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 33: NRIS Results for J1157

J1157 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	29.4	5.39%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	2,417,694	275,000,000
Big Cajun - Webre 500 kV	80.9	13.36%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	2,509,743	275,000,000
Coly - Willow Glen 500 kV	100.0	10.52%	No		J1020, J1049, J1070, J1113, J1130, J1144, J1157, J1184, J964, J988	138,960	3,000,000

9.3.30. J1158

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 34 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 34: NRIS Results for J1158

J1158 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcucu - Frisco 230 kV #1	0.0	12.42%	No		J1076, J1079, J1142, J1158	148,563	2,500,000
Tezcucu - Frisco 230 kV #2	70.0	12.36%	No		J1076, J1079, J1142, J1158	147,845	2,500,000

9.3.31. J1159

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 35 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 35: NRIS Results for J1159

J1159 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Lake Charles Bulk S. - Lowe Grout Rd 138 kV	0.0	5.27%	No		J1120, J1159, J1162, J1168, J1186	2,741,217	186,400,000
J1165 POI - Lake Charles Bulk N. 138 kV	0.0	5.79%	No		J1100, J1120, J1159, J1162, J1168, J1186	3,511,916	195,200,000
Nelson - Penton Road 230 kV	0.0	11.84%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	2,666,137	23,800,000
Cocodrie - Ville Platte 230 kV	0.0	6.56%	No		J1100, J1120, J1159, J1162, J1168, J1186	976,296	175,000,000
J1165 POI - Colonial Welsh Tap 138 kV	0.0	5.79%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,489,979	185,440,000
J1165 POI - Jennings 138 kV	0.0	5.45%	No		J1120, J1159, J1162, J1165, J1168, J1186	3,441,691	204,640,000
Bayou Labutte 500/230 kV autotransformer	0.0	6.10%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,029,473	18,000,000
Sabine - Mud Lake 230 kV	0.0	15.22%	No		J1100, J1120, J1159, J1162, J1168, J1186	10,603,642	81,500,000
Webre - Bayou Labutte 500 kV	0.0	8.50%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,596,767	275,000,000
Richard 500/138 kV autotransformer	0.0	9.21%	No		J1120, J1162, J1168, J1186, J1159, J1100	2,639,479	20,000,000
Ville Platte - West Fork 230 kV	2.3	6.56%	No		J1100, J1120, J1159, J1162, J1168, J1186	976,296	175,000,000
Rork - Sellers Rd 230 kV	6.5	10.87%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	3,224,408	31,000,000
Bayou Cove - Richard 138 kV	11.2	5.20%	No		J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,229,104	57,200,000
Henning - Hebert 138 kV	26.0	5.79%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	951,970	176,500,000

Bayou Labutte - Willow Glen 500 kV	29.0	7.35%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,380,734	275,000,000
Hebert - Bayou Cove 138 kV	30.4	5.79%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	861,701	175,000,000
Colonial Welsh Tap - Henning 138 kV	39.2	5.79%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	861,701	175,000,000
New Long John - Dayton 138 kV	45.5	6.31%	No		J1100, J1159, J1162, J1186	1,450,505	9,600,000
J1166 POI - Richard 138 kV	47.7	5.86%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,385,105	57,200,000
J1166 POI - Bayou Cove 138 kV	56.3	5.86%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,385,105	57,200,000
J683 Tap - Wilbert 138 kV	64.5	5.95%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186, J961	91,457	2,760,000
Huntsville - Wyntx Tap 138 kV	65.2	50.29%	No		J1100, J1159	3,894,839	5,700,000
Nelson 500/230 kV autotransformer	68.5	23.99%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	3,570,328	175,000,000
Champagne - Bobcat 138 kV	68.9	5.95%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	648,611	6,728,000
Rivtrin - Staley 138 kV	72.9	16.26%	No		J1100, J1159	6,468,176	13,000,000
Ponderosa - Conroe 138 kV	73.6	11.41%	No		J1100, J1159	1,288,267	5,100,000
Bayou Labutte - Iberville 230 kV	73.9	6.10%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	897,929	15,700,000
Big Cajun - Fancy Point 500 kV	79.7	16.05%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	3,015,073	275,000,000
J643 Tap - Marshall 138 kV	87.6	5.13%	No		J1120, J1159, J1162, J1168, J1186	4,344,070	28,040,000
Nelson - Richard 500 kV	88.9	30.78%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	4,580,854	175,000,000
Richard - Wells 500 kV	100.0	28.23%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	303,264	3,000,000

9.3.32. J1162

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 36 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 36: NRIS Results for J1162

J1162 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Nelson - Goosport 138 kV	0.0	5.13%	No		J1120, J1162, J1168	610,781	175,000,000
Nelson - Lake Charles Bulk S. 138 kV	0.0	5.14%	No		J1120, J1162, J1168	611,971	175,000,000
Lake Charles Bulk N. - Goosport 138 kV	0.0	5.13%	No		J1120, J1162, J1168	610,781	175,000,000
Lake Charles Bulk S. - Lowe Grout Rd 138 kV	0.0	6.22%	No		J1120, J1159, J1162, J1168, J1186	2,588,292	186,400,000

J1165 POI - Lake Charles Bulk N. 138 kV	0.0	6.83%	No		J1100, J1120, J1159, J1162, J1168, J1186	3,314,182	195,200,000
Nelson - Penton Road 230 kV	0.0	14.86%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	2,676,945	23,800,000
Cocodrie - Ville Platte 230 kV	0.0	6.84%	No		J1100, J1120, J1159, J1162, J1168, J1186	814,374	175,000,000
J1165 POI - Colonial Welsh Tap 138 kV	0.0	6.83%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,406,087	185,440,000
J1165 POI - Lowe Grout Rd 138 kV	0.0	5.77%	No		J1120, J1162, J1168, J1186	2,635,170	184,960,000
J1165 POI - Jennings 138 kV	0.0	6.43%	No		J1120, J1159, J1162, J1165, J1168, J1186	3,248,451	204,640,000
Bayou Labutte 500/230 kV autotransformer	0.0	6.86%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	926,188	18,000,000
Sabine - Mud Lake 230 kV	0.0	18.84%	No		J1100, J1120, J1159, J1162, J1168, J1186	10,500,532	81,500,000
Webre - Bayou Labutte 500 kV	0.0	9.28%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,394,635	275,000,000
Richard 500/138 kV autotransformer	0.0	10.97%	No		J1120, J1162, J1168, J1186, J1159, J1100	2,515,100	20,000,000
Ville Platte - West Fork 230 kV	1.8	6.84%	No		J1100, J1120, J1159, J1162, J1168, J1186	814,374	175,000,000
Rork - Sellers Rd 230 kV	5.2	12.80%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	3,037,529	31,000,000
Bayou Cove - Richard 138 kV	9.0	6.14%	No		J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,161,030	57,200,000
Henning - Hebert 138 kV	20.8	6.83%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	898,371	176,500,000
Bayou Labutte - Willow Glen 500 kV	23.2	8.04%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,208,283	275,000,000
Hebert - Bayou Cove 138 kV	24.3	6.83%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	813,183	175,000,000
Colonial Welsh Tap - Henning 138 kV	31.4	6.83%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	813,183	175,000,000
New Long John - Dayton 138 kV	36.4	13.54%	No		J1100, J1159, J1162, J1186	2,489,996	9,600,000
J1166 POI - Richard 138 kV	38.2	6.91%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,306,632	57,200,000
J1166 POI - Bayou Cove 138 kV	45.0	6.91%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,306,632	57,200,000
J683 Tap - Wilbert 138 kV	52.1	6.73%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186, J961	82,757	2,760,000
Nelson 500/230 kV autotransformer	54.8	28.19%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	3,356,316	175,000,000
Champagne - Bobcat 138 kV	58.9	6.73%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	586,911	6,728,000
Bayou Labutte - Iberville 230 kV	59.1	6.86%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	807,842	15,700,000
Big Cajun - Fancy Point 500 kV	63.8	18.12%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,723,146	275,000,000
J643 Tap - Marshall 138 kV	70.1	6.18%	No		J1120, J1159, J1162, J1168, J1186	4,186,566	28,040,000

Nelson - Richard 500 kV	71.1	35.69%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	4,249,270	175,000,000
Richard - Wells 500 kV	80.0	32.47%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	284,239	3,000,000

9.3.33. J1163

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 37 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 37: NRIS Results for J1163

J1163 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIIS Network Upgrade	Projects Associated with ERIIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	35.2	5.89%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	1,327,759	275,000,000
Big Cajun - Webre 500 kV	112.6	5.61%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	1,264,640	275,000,000
Twinkletown - Freeport 230 kV	120.0	34.37%	No		J1070, J1093, J1095, J1113, J1163, J962	1,553,584	14,312,000

9.3.34. J1165

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 38 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 38: NRIS Results for J1165

J1165 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIIS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIIS Network Upgrade	Projects Associated with ERIIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Manena - Sadie 230 kV	0.0	10.11%	No		J1165, J1166	2,256,941	175,000,000
Richard - Colonial Academy 138 kV	0.0	5.97%	No		J1165, J1166	12,263,195	17,960,000
Nelson - Penton Road 230 kV	0.0	8.36%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	2,823,763	23,800,000
Acadia - Scanlan 138 kV	0.0	5.97%	No		J1165, J1166	2,835,010	4,152,000
J1165 POI - Colonial Welsh Tap 138 kV	0.0	34.72%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	13,402,090	185,440,000
Colonial Academy - Acadia 138 kV	0.0	5.97%	No		J1165, J1166	9,837,868	14,408,000
J1165 POI - Jennings 138 kV	0.0	19.75%	No		J1120, J1159, J1162, J1165, J1168, J1186	18,708,272	204,640,000

Bayou Labutte 500/230 kV autotransformer	0.0	7.93%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,007,473	18,000,000
Webre - Bayou Labutte 500 kV	3.4	9.87%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,781,193	275,000,000
Rork - Sellers Rd 230 kV	9.7	11.80%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	5,250,416	31,000,000
Bayou Cove - Richard 138 kV	16.8	18.50%	No		J1120, J1159, J1162, J1165, J1166, J1168, J1186	6,559,159	57,200,000
Henning - Hebert 138 kV	39.0	34.72%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	8,562,801	176,500,000
Bayou Labutte - Willow Glen 500 kV	43.5	8.58%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,417,694	275,000,000
Hebert - Bayou Cove 138 kV	45.6	34.72%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	7,750,840	175,000,000
Colonial Welsh Tap - Henning 138 kV	68.2	34.72%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	7,750,840	175,000,000
J1166 POI - Richard 138 kV	71.6	20.84%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	7,388,804	57,200,000
J1166 POI - Bayou Cove 138 kV	75.6	20.84%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	7,388,804	57,200,000
L673 Tap - Goosport 69 kV	84.5	8.19%	No		J1165	2,468,000	2,468,000
J683 Tap - Wilbert 138 kV	97.7	8.54%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186, J961	196,902	2,760,000
Nelson 500/230 kV autotransformer	102.8	20.82%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	4,647,825	175,000,000
Champagne - Bobcat 138 kV	103.8	8.54%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,396,421	6,728,000
North Crowley - Scott 138 kV	110.4	5.47%	No		J1165, J1166	19,700,383	28,200,000
Bayou Labutte - Iberville 230 kV	110.8	7.93%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,750,962	15,700,000
Big Cajun - Fancy Point 500 kV	131.5	21.27%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	5,993,514	275,000,000
Nelson - Richard 500 kV	133.3	17.72%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	3,955,786	175,000,000
Richard - Wells 500 kV	150.0	33.34%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	546,842	3,000,000

9.3.35. J1166

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 39 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 39: NRIS Results for J1166

J1166 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIS Network Upgrade	Projects Associated with ERIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Manena - Sadie 230 kV	0.0	5.02%	No		J1165, J1166	373,552	175,000,000
Richard - Colonial Academy 138 kV	0.0	8.32%	No		J1165, J1166	5,696,805	17,960,000
Acadia - Scanlan 138 kV	0.0	8.32%	No		J1165, J1166	1,316,990	4,152,000
Colonial Academy - Acadia 138 kV	0.0	8.32%	No		J1165, J1166	4,570,132	14,408,000
Bayou Labutte 500/230 kV autotransformer	0.0	8.20%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	691,941	18,000,000
Webre - Bayou Labutte 500 kV	1.1	9.70%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	911,097	275,000,000
Rork - Sellers Rd 230 kV	3.3	9.10%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,349,683	31,000,000
Bayou Cove - Richard 138 kV	13.0	30.00%	No		J1120, J1159, J1162, J1165, J1166, J1168, J1186	3,545,491	57,200,000
Bayou Labutte - Willow Glen 500 kV	22.7	8.45%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	793,687	275,000,000
J1166 POI - Richard 138 kV	28.2	34.24%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	4,046,587	57,200,000
J683 Tap - Wilbert 138 kV	34.3	9.41%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186, J961	72,321	2,760,000
Champagne - Bobcat 138 kV	34.6	9.41%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	512,893	6,728,000
North Crowley - Scott 138 kV	36.8	7.08%	No		J1165, J1166	8,499,617	28,200,000
Bayou Labutte - Iberville 230 kV	36.9	8.20%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	603,526	15,700,000
Big Cajun - Fancy Point 500 kV	44.4	22.24%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,088,948	275,000,000
Richard - Wells 500 kV	50.0	30.46%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	163,184	3,000,000

9.3.36. J1168

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 40 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 40: NRIS Results for J1168

J1168 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Nelson - Goosport 138 kV	0.0	5.16%	No		J1120, J1162, J1168	614,352	175,000,000
Nelson - Lake Charles Bulk S. 138 kV	0.0	5.18%	No		J1120, J1162, J1168	616,734	175,000,000
Lake Charles Bulk N. - Goosport 138 kV	0.0	5.16%	No		J1120, J1162, J1168	614,352	175,000,000
Lake Charles Bulk S. - Lowe Grout Rd 138 kV	0.0	6.21%	No		J1120, J1159, J1162, J1168, J1186	2,584,131	186,400,000
J1165 POI - Lake Charles Bulk N. 138 kV	0.0	6.83%	No		J1100, J1120, J1159, J1162, J1168, J1186	3,314,182	195,200,000
Nelson - Penton Road 230 kV	0.0	14.70%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	2,648,122	23,800,000
Cocodrie - Ville Platte 230 kV	0.0	6.84%	No		J1100, J1120, J1159, J1162, J1168, J1186	814,374	175,000,000
J1165 POI - Colonial Welsh Tap 138 kV	0.0	6.83%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,406,087	185,440,000
J1165 POI - Lowe Grout Rd 138 kV	0.0	5.76%	No		J1120, J1162, J1168, J1186	2,630,603	184,960,000
J1165 POI - Jennings 138 kV	0.0	6.43%	No		J1120, J1159, J1162, J1165, J1168, J1186	3,248,451	204,640,000
Bayou Labutte 500/230 kV autotransformer	0.0	6.84%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	923,488	18,000,000
Sabine - Mud Lake 230 kV	0.0	18.24%	No		J1100, J1120, J1159, J1162, J1168, J1186	10,166,120	81,500,000
Webre - Bayou Labutte 500 kV	0.0	9.25%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,390,127	275,000,000
Richard 500/138 kV autotransformer	0.0	10.83%	No		J1120, J1162, J1168, J1186, J1159, J1100	2,483,002	20,000,000
Ville Platte - West Fork 230 kV	1.8	6.84%	No		J1100, J1120, J1159, J1162, J1168, J1186	814,374	175,000,000
Rork - Sellers Rd 230 kV	5.2	12.71%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	3,016,171	31,000,000
Bayou Cove - Richard 138 kV	9.0	6.13%	No		J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,159,139	57,200,000
Henning - Hebert 138 kV	20.8	6.83%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	898,371	176,500,000
Bayou Labutte - Willow Glen 500 kV	23.2	8.01%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,203,775	275,000,000
Hebert - Bayou Cove 138 kV	24.3	6.83%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	813,183	175,000,000
Colonial Welsh Tap - Henning 138 kV	36.4	6.83%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	813,183	175,000,000
J1166 POI - Richard 138 kV	38.2	6.91%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,306,632	57,200,000
J1166 POI - Bayou Cove 138 kV	45.0	6.91%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,306,632	57,200,000

J683 Tap - Wilbert 138 kV	52.1	6.71%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186, J961	82,511	2,760,000
Nelson 500/230 kV autotransformer	54.8	28.14%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	3,350,363	175,000,000
Champagne - Bobcat 138 kV	58.9	6.71%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	585,167	6,728,000
Bayou Labutte - Iberville 230 kV	59.1	6.84%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	805,487	15,700,000
Big Cajun - Fancy Point 500 kV	63.8	18.06%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	2,714,129	275,000,000
J643 Tap - Marshall 138 kV	70.1	6.67%	No		J1120, J1159, J1162, J1168, J1186	4,518,511	28,040,000
Nelson - Richard 500 kV	71.1	35.41%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	4,215,933	175,000,000
Richard - Wells 500 kV	80.0	32.29%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	282,491	3,000,000

9.3.37. J1184

This generator is determined to be deliverable up to 40.5 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 41 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 41: NRIS Results for J1184

J1184 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERI and case assumptions)		40.5 MW (80.92%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERI Network Upgrade	Projects Associated with ERI Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Coly - Willow Glen 500 kV	50.0	9.98%	No		J1020, J1049, J1070, J1113, J1130, J1144, J1157, J1184, J964, J988	41,807	3,000,000

9.3.38. J1186

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 42 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 42: NRIS Results for J1186

J1186 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERI and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERI Network Upgrade	Projects Associated with ERI Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Lake Charles Bulk S. - Lowe Grout Rd 138 kV	0.0	5.96%	No		J1120, J1159, J1162, J1168, J1186	3,100,124	186,400,000
J1165 POI - Lake Charles Bulk N. 138 kV	0.0	6.55%	No		J1100, J1120, J1159, J1162, J1168, J1186	3,972,893	195,200,000

Nelson - Penton Road 230 kV	0.0	14.03%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	3,159,282	23,800,000
Cocodrie - Ville Platte 230 kV	0.0	6.79%	No		J1100, J1120, J1159, J1162, J1168, J1186	1,010,526	175,000,000
J1165 POI - Colonial Welsh Tap 138 kV	0.0	6.55%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,685,555	185,440,000
J1165 POI - Lowe Grout Rd 138 kV	0.0	5.53%	No		J1120, J1162, J1168, J1186	3,156,952	184,960,000
J1165 POI - Jennings 138 kV	0.0	6.17%	No		J1120, J1159, J1162, J1165, J1168, J1186	3,896,373	204,640,000
Bayou Labutte 500/230 kV autotransformer	0.0	6.67%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,125,670	18,000,000
Sabine - Mud Lake 230 kV	0.0	17.59%	No		J1100, J1120, J1159, J1162, J1168, J1186	12,254,801	81,500,000
Webre - Bayou Labutte 500 kV	0.0	9.08%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,705,723	275,000,000
Richard 500/138 kV autotransformer	0.0	10.52%	No		J1120, J1162, J1168, J1186, J1159, J1100	3,014,910	20,000,000
Ville Platte - West Fork 230 kV	2.3	6.79%	No		J1100, J1120, J1159, J1162, J1168, J1186	1,010,526	175,000,000
Rork - Sellers Rd 230 kV	6.5	12.30%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	3,648,594	31,000,000
Bayou Cove - Richard 138 kV	11.2	5.89%	No		J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,392,196	57,200,000
Henning - Hebert 138 kV	26.0	6.55%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,076,927	176,500,000
Bayou Labutte - Willow Glen 500 kV	29.0	7.86%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,476,540	275,000,000
Hebert - Bayou Cove 138 kV	30.4	6.55%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	974,808	175,000,000
Colonial Welsh Tap - Henning 138 kV	39.2	6.55%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	974,808	175,000,000
New Long John - Dayton 138 kV	45.5	14.18%	No		J1100, J1159, J1162, J1186	3,259,614	9,600,000
J1166 POI - Richard 138 kV	47.7	6.63%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	1,567,107	57,200,000
J1166 POI - Bayou Cove 138 kV	56.3	6.63%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	1,567,107	57,200,000
J683 Tap - Wilbert 138 kV	65.2	6.53%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186, J961	100,373	2,760,000
Nelson 500/230 kV autotransformer	66.9	27.06%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	4,027,222	175,000,000
Coley Creek - Shepherd 138 kV	68.5	99.90%	Yes	J1186	J1186	0	4,520,000
Champagne - Bobcat 138 kV	73.6	6.53%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	711,837	6,728,000
Bayou Labutte - Iberville 230 kV	73.9	6.67%	No		J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	981,834	15,700,000
Big Cajun - Fancy Point 500 kV	79.7	17.60%	Yes	J1073	J1073, J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	3,306,248	275,000,000
J643 Tap - Marshall 138 kV	87.6	5.90%	No		J1120, J1159, J1162, J1168, J1186	4,996,104	28,040,000

Nelson - Richard 500 kV	88.9	34.43%	No		J1100, J1120, J1159, J1162, J1165, J1168, J1186	5,124,067	175,000,000
Richard - Wells 500 kV	100.0	31.38%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186	341,978	3,000,000

9.3.39. J961

This generator is determined to be deliverable up to 84.5 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 43 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 43: NRIS Results for J961

J961 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)				84.5 MW (56.30%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIS Network Upgrade	Projects Associated with ERIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
J683 Tap - Wilbert 138 kV	147.5	78.56%	No		J1100, J1120, J1159, J1162, J1165, J1166, J1168, J1186, J961	1,811,318	2,760,000
Jaguar - Esso Tap 230 kV	150.0	9.98%	No		J961	6,640,000	6,640,000

9.3.40. J962

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 44 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 44: NRIS Results for J962

J962 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIS Network Upgrade	Projects Associated with ERIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	29.4	5.35%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	1,005,024	275,000,000
Big Cajun - Webre 500 kV	93.9	5.00%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	939,275	275,000,000
Twinkletown - Freeport 230 kV	100.0	77.79%	No		J1070, J1093, J1095, J1113, J1163, J962	2,930,203	14,312,000

9.3.41. J964

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 45 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 45: NRIS Results for J964

J964 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	20.6	5.68%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	1,788,380	275,000,000
Big Cajun - Webre 500 kV	56.6	14.18%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	1,864,649	275,000,000
Coly - Willow Glen 500 kV	70.0	11.13%	No		J1020, J1049, J1070, J1113, J1130, J1144, J1157, J1184, J964, J988	102,912	3,000,000

9.3.42. J972

This generator is determined to be deliverable up to 64.1 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 46 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 46: NRIS Results for J972

J972 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)				64.1 MW (45.76%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERS Network Upgrade	Projects Associated with ERS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Perryville - Baxter Wilson 500 kV	82.1	5.93%	No		J1007, J1023, J1060, J1091, J1093, J1097, J1125, J1137, J1147, J1156, J972	142,802	3,000,000
West Memphis 500/161 kV autotransformer	112.6	48.33%	No		J1007, J1023, J1078, J1091, J1125, J1137, J1146, J1147, J1155, J1156, J972	2,656,761	15,000,000
J1125 POI - Wynne E. 161 kV	122.6	38.05%	No		J1125, J1147, J972	1,557,693	5,000,000
Marked Tree - Harrisburg 161 kV	123.9	27.49%	No		J1007, J1023, J1060, J1125, J1147, J1156, J972	507,901	3,600,000
Hughes - West Memphis 161 kV	128.9	59.80%	No		J1125, J1147, J1156, J972	17,394,498	28,000,000
J1023 POI - Osceola 161 kV	132.5	14.24%	No		J1007, J1023, J1125, J1147, J1156, J972	57,612	500,000
Moses - Forest City North 161 kV	134.0	38.05%	No		J1147, J972	3,109,023	5,000,000
Wynne E. - Parkin 161 kV	140.0	38.05%	No		J1125, J1147, J972	4,766,542	15,300,000

9.3.43. J988

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 47 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 47: NRIS Results for J988

J988 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		0.0 MW (0.0%)					
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Associated with ERIS Network Upgrade	Projects Associated with ERIS Network Upgrade	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
McKnight - Coly 500 kV	58.7	8.78%	Yes	J1073	J1020, J1049, J1070, J1073, J1078, J1093, J1095, J1113, J1130, J1144, J1146, J1155, J1157, J1163, J962, J964, J988	3,298,734	275,000,000
Big Cajun - Webre 500 kV	161.8	9.09%	Yes	J1073	J1020, J1049, J1070, J1093, J1095, J1113, J1130, J1144, J1157, J1163, J962, J964, J988	3,415,204	275,000,000
Coly - Willow Glen 500 kV	200.0	8.41%	No		J1020, J1049, J1070, J1113, J1130, J1144, J1157, J1184, J964, J988	222,177	3,000,000

10. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis, which tests for Network Upgrades driven by higher queued interconnection projects, was performed for this System Impact Study.

The maximum MW impacts and SNU cost allocations appear in Table 48.

Table 48: Maximum MW Impact and SNU Cost Allocations

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility (\$)
Nelson – Goosport – Lake Charles Bulk N. 138 kV Rebuild	DPP-2016-AUG	J484	49.90	11,400,000	6,346,154
	DPP-2018-APR	J1165	39.74		5,053,846
Kuhn Rd – Ebony 161 kV Rebuild	DPP-2017-AUG	J934	133.02	4,900,000	2,380,760
	DPP-2018-APR	J1007	16.31		291,841
	DPP-2018-APR	J1060	13.31		238,219
	DPP-2018-APR	J1125	23.18		414,813
	DPP-2018-APR	J1156	87.96		1,574,367

11. Cost Allocation

The cost allocation of Network Upgrades (NU) for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of NRIS as of the draft System Impact Study report date.

11.1. Cost Assumptions for Network Upgrades

The cost estimate for each NU identified in the System Impact Study is provided by the corresponding Transmission Owner.

11.2. Cost Allocation Methodology

The costs of NU for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all study group generation projects dispatched in the Study Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any Network Upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. The cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Project A Cost Portion of NU} = \frac{\text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.



Appendix A – Cost Allocation Summary (CEII)

Appendix B – Network Upgrade Table Per Project (CEII)

Appendix C – MISO ERIS Analysis (CEII)

Appendix D – MISO Deliverability Analysis (CEII)